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Assignment - 3
Section - A

1. What is SQL? Why is it important?
→ SQL is used to store, retrive and manage data relational database. It is important becoz it allows us interact with databases efficiently, perform operations inserting, updating, deleting & retrieving data.

2. Difference b/w DDL & DML?
→ DDL: Used to define structure (create, Alter, Dro
DML: used to manipulate data (insert, update, delete)

3. What is a Join? Types
→ Join is used to combine data from multiple based on a common coloumn.
Types :-
Inner Join : Only matching records
· Left Join : all left + matching right
Right Join : all right + matching left
Full Join : all records from both tables.

4. Group by & Having
→ Group By : Groups rows with same values.
Having : Filters grouped data.

Eg → Count students per department and filter House >

5. What is Subquery?
Ans. A subquery is a query inside another query used to complex operations.

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. What is Relational Algebra?
: It is a procedural query language used in DBMS with operations like: select, Project, Join

Where vs Having?
Where → filters before grouping
Having → filters after grouping

Section - B

Students with marks > 75
, Select * from students
Where marks > 75;

Join Query (student + course)
, Select S.name, C.course_name
From Students S
Inner Join Courses C
On S.course_id = C.Course_id;

) Count students per department
, Select department, count (*) As total_students
from Students
Group By departments;

Section - C

. Employees in specific department
→ Select 1.name
from Employees
Join Departments_id

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on e.depart_id = d.depart_id
where d.depart_name = 'HR';

2. Total Sales per Customer
→ Select customer_id, SUM(amount) As total_sales
from Orders
Group By ~~cus~~ customer_id;

3. Top 3 students
→ Select * from students
Order by marks Desc
Limit 3;

Section - D
Example Query (Real-life : online store)

→ Select product_id, sum(quantity) As total_sold from Sales
Group by product_id;

* This helps business identify best_selling products.

Join use Case

Select o.order_id, c.customer_name
from order o
Join Customers c
On o.customer_id = c.customer_id;

* used to link related data across tables.

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Group By Explanation

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Select department , AVG(Salary)
from Employees
Group By department;
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* Output shows average salary per department.